UNICEF CLIMATE VENTURES 2026

PROGRAMME APPLICATION DOCUMENT

Climate-Resilient School Infrastructure Toolkit (CRIT)

Empowering Communities to Build Safer, Sustainable Schools for Children

Submitted by: WUM Advisory Services

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1. EXECUTIVE SUMMARY

Pakistan's children are among the most exposed in the world to climate-driven shocks that directly threaten their health, safety, and right to uninterrupted education. Recurring floods, intensifying heatwaves, drought stress, unsafe drinking water systems, and structural deterioration of public infrastructure continue to disrupt learning environments at scale. These shocks disproportionately impact girls, children with disabilities, and children in low-income rural communities, reinforcing systemic exclusion and increasing risks of dropout, child labour, and early marriage.

The Climate-Resilient School Infrastructure Toolkit (CRIT) is a women-led, open source programme designed to protect children at the most critical frontline of service delivery: the school. CRIT delivers a scalable and locally adaptable model to strengthen school infrastructure through low-cost retrofitting, community-led training, inclusive design upgrades, and climate monitoring systems. The programme is structured to reduce reliance on contractor-heavy models and imported engineering designs that are often unaffordable, slow to deploy, and difficult to sustain at scale.

CRIT aligns directly with the strategic priorities of UNICEF Climate Ventures 2026, including child-centred innovation, climate and health resilience, inclusive design, open source knowledge systems, and climate-informed public service strengthening. CRIT will pilot and validate practical retrofitting packages, integrate hazard mapping and vulnerability analytics into school planning, and publish a replicable toolkit for adoption by provincial education systems and partner organizations.

The programme will focus implementation in climate high-risk districts across Sindh, Punjab, and Balochistan, where climate exposure intersects with fragile infrastructure and high learning poverty.

Key Results Expected (Year 1)

By the end of the 12-month programme cycle, CRIT will deliver measurable and verifiable outcomes, including:

- **50 schools retrofitted** using CRIT resilience standards in climate-exposed locations.
- **1,000 stakeholders trained**, including teachers, school management committee members, parents, students, and local governance representatives.
- **Deployment of low-cost monitoring tools** to track heat stress, structural risks, and water safety in targeted schools.

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- **Publication of the open source CRIT Toolkit**, including technical blueprints, DIY retrofitting manuals, training modules, inclusion checklists, and replication guides.
- **Reduction in climate-related school closures by at least 30 percent** across pilot schools through targeted interventions and preparedness planning.
- **Strengthened policy uptake pathway**, supported by evidence products, technical briefings, and structured engagement with provincial education departments.

Total Requested Funding

USD 93,500 (fully compliant with the UNICEF Climate Ventures ceiling of USD 100,000)

Duration

12 Months (September 2026 to August 2027)

CRIT offers UNICEF a scalable, high-value innovation that strengthens education continuity and child protection at the climate-education-health nexus, while generating open source outputs suitable for national expansion and global replication.

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2. COUNTRY CONTEXT AND SECTOR OVERVIEW

2.1 Climate Change and Education in Pakistan

Pakistan ranks among the most climate-vulnerable countries globally, facing escalating floods, drought cycles, extreme heat events, windstorms, and irregular rainfall patterns. Climate shocks are no longer occasional disruptions. They have become structural threats to public service delivery and child wellbeing.

The education sector has suffered significant damage. The 2022 floods alone reportedly damaged or destroyed more than **27,000 schools**, displacing millions of children and interrupting learning for prolonged periods. In many flood-affected districts, school rehabilitation remains incomplete due to funding constraints, slow procurement processes, and limited technical capacity.

Beyond flooding, **heatwaves increasingly disrupt school attendance** and learning performance. High temperatures reduce cognitive function, increase dehydration risks, and contribute to higher absenteeism. In parallel, unsafe water supply systems, weak sanitation infrastructure, and deteriorating school buildings increase illness rates and exposure to protection risks.

Despite clear urgency, education-sector climate adaptation remains fragmented. Existing approaches are often centralized, contractor-driven, and costly. These models rarely generate local ownership or sustainable maintenance systems.

2.2 Education Equity Gaps and Climate Exposure

Climate shocks intensify existing inequalities in Pakistan. Vulnerable children experience disproportionate disruption due to poverty, gender exclusion, disability barriers, and rural isolation.

- **Girls face heightened barriers** when sanitation facilities are inadequate, privacy is compromised, or safety risks increase during climate events. Poor menstrual hygiene infrastructure directly reduces attendance for adolescent girls.
- **Children with disabilities are frequently excluded** due to inaccessible classrooms, absence of ramps, lack of adapted WASH facilities, and non-inclusive emergency evacuation protocols.
- **Poor and rural communities are hit hardest**, as they possess the weakest recovery resources. These communities also face higher dropout rates following disasters,

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particularly where schools are converted into shelters or remain damaged for long periods.

Climate vulnerability therefore acts as a multiplier of education inequality and exclusion.

2.3 Alignment with National and Provincial Priorities

CRIT is aligned with Pakistan's education and disaster preparedness priorities by supporting:

- School safety and disaster risk reduction planning
- Climate resilience infrastructure improvements
- Inclusive education and disability accessibility
- Strengthened community participation in education governance
- Evidence-based planning and prioritization for school rehabilitation

CRIT is designed to complement provincial education reforms by delivering practical and measurable results while producing scalable tools that can be integrated into government planning and implementation systems.

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3. PROBLEM ANALYSIS AND SYSTEM BOTTLENECKS

Pakistan's school infrastructure system faces persistent weaknesses that undermine resilience, preparedness, and inclusion.

3.1 Core System Bottlenecks

Fragile and Unsafe Infrastructure

A large share of public schools operate in buildings that are structurally weak, poorly maintained, and exposed to heat stress and flooding. Roof insulation is often absent, drainage systems are inadequate, and classrooms lack ventilation.

Limited Technical Knowledge for Low-Cost Retrofitting

School communities and local authorities lack accessible guidance on climate-resilient retrofitting using affordable local materials. As a result, schools remain dependent on expensive engineering models and centralized rehabilitation schemes.

Weak Preparedness and Reactive Emergency Capacity

Most schools lack operational disaster preparedness plans, evacuation drills, heat safety protocols, and structured response systems. Coordination with early warning systems is minimal.

Climate Data Blind Spots in Education Planning

Education sector planning does not systematically integrate hazard mapping, risk analytics, and climate exposure scoring into budgeting and prioritization. This results in inefficient resource allocation and delayed response.

Exclusionary Design and Weak Accessibility Standards

Many schools remain unsafe or unusable for girls and children with disabilities. Lack of accessible toilets, ramps, safe pathways, and privacy features reinforces systemic exclusion.

Weak Monitoring and Accountability Systems

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There is limited routine monitoring of school safety indicators. Infrastructure performance is rarely tracked using evidence-based indicators, which weakens accountability and policy decision-making.

CRIT is designed to directly address these bottlenecks through a toolkit-based approach that builds capacity, strengthens governance systems, and produces open source solutions.

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4. PROGRAMME RATIONALE AND STRATEGIC ALIGNMENT

4.1 Why CRIT is Essential

Climate resilience in education is not only an infrastructure issue. It is fundamentally linked to child rights, public health, learning continuity, and long-term human capital development.

Schools are often the first public structures to fail during climate shocks. They are also frequently used as temporary shelters during displacement, which increases risks of overcrowding, unsafe water, disease transmission, and child protection incidents. Strengthening schools is therefore essential for both education continuity and emergency preparedness.

CRIT provides a cost-effective pathway to:

- Improve physical school safety and usability
- Reduce heat-related health risks
- Strengthen WASH resilience
- Reduce climate-driven learning disruptions
- Institutionalize preparedness and maintenance capacity

4.2 Alignment with UNICEF Climate Ventures 2026

CRIT directly supports UNICEF Climate Ventures objectives by delivering:

- Child-centred resilience solutions focused on safe learning environments
- Open source toolkit outputs designed for replication and adoption
- Integration of hazard mapping and anticipatory planning
- Data-driven monitoring and evidence generation
- Embedded gender equality and disability inclusion standards
- Practical innovations suitable for scaling within government systems

CRIT addresses the climate-health-education intersection by strengthening physical safety, reducing heat exposure, improving WASH resilience, and minimizing closures.

4.3 CRIT Innovation Value Proposition

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CRIT offers a unique innovation model by combining physical retrofits with institutional and community systems strengthening.

Key innovation features include:

- Hyperlocal hazard mapping and vulnerability scoring
- Practical retrofitting standards using local materials and labour
- Inclusive infrastructure standards for girls and children with disabilities
- Low-cost monitoring tools and dashboards for decision-making
- Women-led cooperative implementation models
- Fully open source outputs classified as Digital Public Goods

CRIT is therefore not only a set of infrastructure upgrades. It is a replicable resilience system.

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5. PROGRAMME OBJECTIVES

5.1 Overall Objective

To strengthen climate resilience of school infrastructure in Pakistan's high-risk regions through community-led, gender-responsive, open source retrofitting and preparedness systems, ensuring safe and uninterrupted learning for all children.

5.2 Specific Objectives

1. Develop, pilot, and validate the Climate-Resilient School Infrastructure Toolkit (CRIT) in selected high-risk schools across Sindh, Punjab, and Balochistan.
2. Build the capacity of at least 1,000 stakeholders, including teachers, parents, students, school committees, and local authorities, in climate resilience, preparedness, and infrastructure maintenance.
3. Deploy low-cost monitoring and reporting tools that support school-level risk management and evidence-based planning.
4. Reduce climate-related school closures by at least 30 percent across pilot sites through targeted upgrades and preparedness measures.
5. Catalyze policy integration by producing evidence products, technical guidelines, and adoption pathways for provincial education system uptake.

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6. THEORY OF CHANGE AND PROGRAMME LOGIC

6.1 Theory of Change Statement

If climate-exposed schools are equipped with locally adaptable retrofitting tools, inclusive infrastructure standards, and community-led preparedness systems supported by monitoring technology, then schools will remain safer and operational during climate shocks. This will reduce closures, improve child health and protection outcomes, and sustain learning continuity for the most vulnerable children.

6.2 Results Chain

Inputs

UNICEF funding, WUM technical leadership, partnerships with government and universities, engagement of women-led cooperatives, community participation, and open source technical development.

Activities

Participatory risk mapping, toolkit development, retrofitting works, inclusive WASH upgrades, disability accessibility improvements, training workshops, preparedness drills, monitoring system deployment, and policy engagement.

Outputs

50 schools retrofitted, 1,000 stakeholders trained, monitoring dashboard operational, open source toolkit published, and policy engagement package delivered.

Outcomes

Reduced closures, improved safety and preparedness, strengthened gender and disability inclusion, and improved climate-informed education planning.

Long-Term Impact

A more resilient education system that protects children’s right to safe learning environments under increasing climate stress.

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7. PROGRAMME COMPONENTS AND BUDGETED ACTIVITIES

Component 1: Climate Risk Mapping and Vulnerability Assessment (USD 15,000)

Activities include:

- GIS-based hazard mapping and multi-hazard exposure profiling
- School vulnerability scoring and prioritization index development
- Community consultations for validation and ground-truthing
- Establishment of risk classification criteria aligned with education planning needs

Deliverables: district-specific vulnerability maps and a validated list of prioritized schools.

Component 2: Retrofitting and Infrastructure Resilience Upgrades (USD 40,000)

Activities include:

- Flood resilience measures including barriers, raised platforms, and drainage improvement
- Heat mitigation interventions including roof insulation and passive ventilation
- Water storage improvements and WASH resilience upgrades
- Construction of disability-friendly access improvements including ramps and accessible toilets

Deliverables: 50 schools upgraded with tailored resilience packages.

Component 3: Gender-Responsive Training and Community Capacity Building (USD 15,000)

Activities include:

- Modular training sessions for teachers, students, and school committees
- Community-led disaster preparedness planning
- Evacuation drills and heat safety protocol adoption
- Mobilization of women-led groups for monitoring and maintenance leadership

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Deliverables: 1,000 trained stakeholders and preparedness plans for all pilot schools.

Component 4: Monitoring and Open Source Data Platform (USD 10,000)

Activities include:

- Deployment of low-cost IoT sensors for heat, humidity, and water quality
- Development of dashboards and simple reporting tools
- Training on data usage, ethical handling, and basic risk interpretation

Deliverables: functioning monitoring dashboard and alert system for targeted schools.

Component 5: Policy Advocacy and Scaling Strategy (USD 13,500)

Activities include:

- Development of technical guidelines and policy briefs
- Provincial stakeholder roundtables for uptake dialogue
- Documentation of case studies and toolkit dissemination strategy
- Open source repository launch aligned with Digital Public Goods standards

Deliverables: policy engagement package and public release of CRIT toolkit.

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8. IMPLEMENTATION STRATEGY AND APPROACH

CRIT will follow a phased implementation model to ensure learning, quality assurance, and adaptive management.

Phase 1: Research and Toolkit Design (Months 1 to 3)

- Stakeholder consultations and baseline infrastructure assessments
- Development of CRIT toolkit prototype and draft modules
- Establishment of provincial technical working groups for validation

Phase 2: Pilot Retrofitting and Implementation (Months 4 to 9)

- Final selection of 50 schools with education departments
- Execution of retrofits using local labour and materials
- Delivery of training sessions and preparedness drills
- Installation of monitoring sensors and activation of dashboard

Phase 3: Evidence Generation and Policy Uptake (Months 10 to 12)

- Final evaluation and evidence consolidation
- Finalization and publication of CRIT toolkit
- Policy engagement roundtables and integration dialogue
- Open source dissemination for replication and scaling

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9. TARGETING STRATEGY AND BENEFICIARIES

9.1 Geographic Focus

CRIT targets regions where climate risk is highest and education infrastructure vulnerability is most severe:

- **Sindh:** flood-prone districts in the Indus basin
- **Punjab:** districts experiencing extreme heatwaves and urban heat stress
- **Balochistan:** districts impacted by drought and chronic water scarcity

9.2 Beneficiary Profile

Direct Beneficiaries

- 50 schools including students, teachers, and staff
- 1,000 trained stakeholders including teachers, parents, students, and local governance actors

Indirect Beneficiaries

- Approximately 10,000 community members benefiting from safer shelter capacity and improved preparedness systems

9.3 Priority Groups

- Girls and adolescent girls
- Children with disabilities
- Rural and low-income communities
- Schools with repeated climate disruption history

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10. GOVERNANCE AND MANAGEMENT ARCHITECTURE

CRIT will be governed through a structured accountability system designed for transparency and results delivery.

10.1 Steering Committee

Members: UNICEF Pakistan, Ministry and Provincial Education Representatives, WUM leadership, women-led cooperative representatives.

Functions: approval of workplans and budgets, oversight of safeguarding compliance, strategic direction, and monitoring of results.

10.2 Technical Working Groups

Working groups will provide quality assurance and inclusion compliance.

- Climate resilience and engineering standards
- Gender equality and social inclusion
- Disability inclusion and universal accessibility

10.3 Programme Management Unit (PMU)

Led by WUM Advisory Services.

Responsible for daily coordination, procurement tracking, financial reporting, field oversight, documentation, and progress reporting.

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11. MONITORING, EVALUATION, AND LEARNING (MEL)

CRIT will implement a results-based management approach integrating quantitative and qualitative evidence streams.

11.1 Key Indicators (Year 1)

- Schools retrofitted to CRIT standards (Target: 50)
- Stakeholders trained (Target: 1,000)
- Reduction in climate-related closures (Target: 30 percent)
- Proportion of schools with girl-friendly WASH and safety features
- Proportion of schools with disability access upgrades
- Toolkit downloads and repository engagement
- Schools equipped with monitoring sensors (Target: minimum 20)
- Policy engagements conducted (Target: 6)

11.2 Reporting Schedule

- Monthly narrative and financial reports
- Mid-term review at Month 6
- Final evaluation at Month 12
- Third-party verification of selected outputs as required

IoT data will strengthen evidence integrity through real-time monitoring and continuous validation.

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12. BUDGET SUMMARY (USD 93,500)

Category	**Amount (USD)**	**Notes**
Research, Design, and Baseline	15,000	Mapping, assessments, toolkit drafting
Pilot Retrofitting and Implementation	40,000	Minor civil works, materials, upgrades
Monitoring, Evaluation, and Learning	10,000	Sensors, monitoring tools, reporting
Scaling, Advocacy, and Dissemination	20,000	Policy briefs, documentation, open source dissemination
Contingency	8,500	Buffer for inflation and unforeseen costs
Total	93,500	Within UNICEF ceiling

The programme ensures value for money through local sourcing, standardized retrofit modules, open source outputs, and community-driven implementation.

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13. RISK MANAGEMENT AND SAFEGUARDING

CRIT integrates UNICEF safeguarding standards including SEAH prevention, child protection compliance, fiduciary controls, and data privacy protocols.

13.1 Key Risks and Mitigation

- **Low community engagement:** participatory design workshops and local champions
- **Policy resistance:** early alignment with departments and continuous engagement
- **Supply chain disruptions:** diversified vendors and local procurement priority
- **Data privacy risks:** anonymization protocols, encryption, restricted access
- **Safeguarding risks:** code of conduct enforcement, confidential reporting mechanisms, survivor-centred SOPs

Safeguarding tools and complaint pathways will be integrated into all partnership agreements and monitored through compliance audits.

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14. SUSTAINABILITY AND SCALE-UP STRATEGY

CRIT is designed for sustainability through institutional integration, financial expansion pathways, and community ownership.

14.1 Institutional Sustainability

- Integration into provincial infrastructure guidelines
- Capacity transfer to education departments and school committees
- Establishment of routine school safety monitoring practices

14.2 Financial Sustainability

- Provincial co-financing engagement
- CSR and private sector material partnerships
- Replication enabled through open source cost elimination

14.3 Community Ownership

Women-led cooperatives and school committees will be trained to maintain upgrades, conduct safety checks, and lead preparedness drills.

14.4 Scale-Up Roadmap

- Year 1: pilot in 50 schools
- Year 2: expansion to 200 schools
- Year 3 onwards: provincial mainstreaming and national adoption, with global replication potential

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15. CONCLUSION

The Climate-Resilient School Infrastructure Toolkit (CRIT) is a high-impact innovation designed to protect children's learning environments at the frontline of climate risk. It addresses the interconnected challenges of infrastructure fragility, heat stress, flooding exposure, WASH insecurity, gender inequality, and disability exclusion through a practical and scalable solution.

CRIT provides UNICEF with an evidence-based model that is locally owned, cost-efficient, community-driven, and open source. By investing in CRIT, UNICEF will enable a replicable solution that strengthens education continuity, child protection, and climate adaptation capacity in Pakistan and beyond.

WUM Advisory Services welcomes the opportunity to partner with UNICEF Climate Ventures 2026 to deliver this critical intervention for children's safety, learning, and resilience.

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